

NISM VIII

CHAPTER 2: UNDERSTANDING INDEX

LEARNING OBJECTIVES

- ▶ The Index and its significance
- ▶ Different types of stock market indices
- ▶ Index management and maintenance
- ▶ Applications of indices

2.1 & 2.6 INTRODUCTION TO AN INDEX

What is an Index?

- **Collection of securities (shares, bonds, etc)** representing a particular market segment.
- Helps in measuring the changes in an economy or specific sectors.



To know more about various broad market indices

[CLICK HERE](#)

Calculation–

- Each index has its own calculation & methodology, usually expressed in terms of a change from a **base value**.
- If the base value is 100 and the index value today is 110, the index has **increased by 10%**.



2.2 SIGNIFICANCE OF THE STOCK INDEX

Role of an index

- **Measurement:** Tracks price movements of stocks, bonds, etc.
- **Benchmarking:** Used by investors to compare performance.
- **Market Sentiment:** Reflects the overall market behaviour.
- **Underlying for derivatives:** Used as the foundation/base for creating derivatives

2.3 TYPES OF STOCK MARKET INDICES

Market capitalization weighted index



- Weight of each stock based on the **market capitalization**
- Higher the market capitalization of stock higher its weight
- $\text{Market capitalization} = \text{Total outstanding shares} * \text{price per share}$

Example –  Nifty50

EXAMPLE OF MARKET CAPITALIZATION WEIGHTED INDEX

Index value on start date
(1st April, 2023)

Stock	Price (Rs)	Shares (Lakhs)	Market Cap (Rs Lakhs)
AZ	100	10	1000
BY	200	15	3000
CX	150	20	3000
DW	50	25	1250
EU	80	10	800
Total	-	-	9050

Index value today
(1st April, 2024)

Stock	Price (Rs)	Shares (Lakhs)	Market Cap (Rs Lakhs)
AZ	400	10	4000
BY	600	15	9000
CX	500	20	10000
DW	300	25	7500
EU	250	10	2500
Total	-	-	33000

NEW INDEX VALUE CALCULATION

- Old Market Cap: 9050 lakhs
- New Market Cap: 33000 lakhs
- New Index Value $33,000/9,050 = 364.64$
- % return = $(364.64-100)/100(\text{base at the start}) \times 100\% = 264.64\%$



Free Float weighted Index

- Free Float - Excludes shares held by promoters, institutions, etc., focusing only on **shares available for trading**
- Calculations similar to the Market cap method - the only change is free float market cap is considered
- Index is computed based on **weights** of each security's free float market cap

Price weighted Index

- Such an index where each stock's impact on the index is **proportional** to its price. Higher-priced stocks have more influence on the index's performance
- Calculation Method
 - Add the prices of all stocks in the index.**
 - Divide the sum by the total number of stocks in the index.**
- Examples: Dow Jones Industrial Average (DJIA), Nikkei 225



EXAMPLE OF PRICE WEIGHTED INDEX

Old Index (April 1st, 2023)

Stock	Price (Rs)	Shares (Lakhs)
AZ	150	20
BY	300	12
CX	450	16
DW	100	30
EU	250	8
Total	-	-

Old Index Value:

$$(150 + 300 + 450 + 100 + 250)/5 \\ = 1250/5 = 250$$

Index today (April 1st, 2024)

Stock	Price (Rs)	Shares (Lakhs)
AZ	650	20
BY	450	12
CX	600	16
DW	350	30
EU	500	8
Total	-	-

Current Index Value:

$$(650 + 450 + 600 + 350 + 500)/5 \\ = 2550/5 = 510$$

Percentage change

$$= (510 - 250)/250 * 100 = 104\%$$

Equal weighted Index

- Regardless of its market price or market capitalization
- Each stock in the index is assigned the **same weight**.
- The index **requires periodic rebalancing** to maintain equal weights.
- Stocks that increase in price are sold, and stocks that decrease in price are bought.

EXAMPLE OF EQUAL WEIGHTED INDEX

Consider an index constructed on April 2023 , with four stocks (P, Q, R, S). Each stock has the same initial value.

START OF INDEX

STOCK	PRICE (RS) - APRIL 2023	QUANTITY	VALUE(RS) = PRICE * QUANTITY	WEIGHT - APRIL 2023
P	120	250	30,000	0.25
Q	140	214	30,000	0.25
R	160	188	30,000	0.25
S	180	167	30,000	0.25
Total	-	-	1,20,000	1.00

POST 1 YEAR

STOCK	PRICE (RS) - APRIL 2024	QUANTITY (SAME)	CURRENT VALUE(RS) - QTY* PRICE	WEIGHT - APRIL 2023	PRICE CHANGES	RETURN - PRICE CHANGES * OLD WEIGHT
P	170	250	42,500	0.325	41.67%	10.42%
Q	110	214	23,540	0.18	-21.43%	-5.36%
R	210	188	39,480	0.305	31.25%	7.81%
S	150	167	25,050	0.19	-16.67%	4.17%
Total	-	-	1,30,570	1.00	-	8.70%

Index Value Calculation

- Old Index Value:** Set to 100.
- New Index Value=** $(130570/120000) * 100 = 108.81$

Rebalancing

- Increased Weight:** Stocks P and R have weights greater than 25% due to price increases.
- Decreased Weight:** Stocks Q and S have weights less than 25% due to price decreases.
- Action:** Sell quantities of stocks P and R, and buy quantities of stocks Q and S to restore equal weights.



Example - U.S. businesses - Equal-Weighted Wilshire Large-Cap Index

Want to learn more on Equal weighted Index funds (Indian Context) -

[CLICK HERE](#)



2.4 ATTRIBUTES OF AN INDEX

- Reflect the targeted market behaviour
- Should be bias free & computed by **independent third party**



Impact Cost – **additional cost** incurred when executing large orders due to price changes from the ideal price.

1. **Ideal Price:** The average of the best buy and sell prices. For example, if the best bid is Rs. 4.00 and the best offer is Rs. 4.50, the ideal price is Rs. 4.25.

ORDER BOOK

QUANTITY	BID PRICE (IN RS)	OFFER PRICE (IN RS.)	SELL QUANTITY	SR. NO.
1000	4.00	4.50	2000	5
1000	3.90	4.55	1000	6
2000	3.80	4.70	500	7
1000	3.70	4.75	100	8

2. **Actual Price:** The price at which a transaction occurs. When buying, it's higher than the ideal price; when selling, it's lower.

3. Impact Cost Calculation:

- **Ideal price:** (Best Buy + Best Sell) / 2
- **Actual Price:** Weighted average of prices for the order size.
- **Impact Cost** = (Actual Price – Ideal Price)/ideal price × 100



2.5 INDEX MANAGEMENT



Index Construction: Choosing stocks for the index and deciding the calculation methodology.

Index Maintenance: Adjusting the index for corporate actions like bonus issues, rights issues, stock splits, consolidations, and mergers.

Index Revision: Replacing existing stocks with new ones due to changes in trading patterns or market interest.



All 3 activities done by agencies – BSE indices are managed by **Asia Index Pvt Ltd** and NSE indices are managed by **NSE Indices Limited**.

2.6 APPLICATION OF INDICES

Apart from measuring the market sentiments Index is used for these purposes –

Index Funds

- Aim to generate returns **equivalent to a specific index** by investing in the same stocks in the same proportions as the index.
(Example – UTI Nifty 50 Index fund)
- Funds experience slight deviations known as "**tracking errors**" due to fund management expenses and cash holdings for redemptions.



Index Derivatives

- Derivative contracts with an index as the underlying asset, used for hedge/ speculating
- Example – S&P BSE SENSEX 50



ETF

- Replicate **index return** & trade like individual stocks on exchanges.
- Allow for intraday trading.
- Offer **smaller denominations** and low transaction costs.